

# Climate Change and AI: From Power Demand to Productivity

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Climate and environmental investment today sit at the crossroads of three structural shifts: artificial intelligence (AI), deglobalisation and electrification. Each is creating demand for physical assets, new supply chains and higher levels of capital spending. This is the first in a series looking at how these megatrends are reshaping the opportunities for climate-focused investors, beginning with AI.

Expected demand for artificial intelligence is driving one of the largest and most expensive infrastructure buildouts in living memory. The supply chain impacts are broad and deep, overlapping with many areas that until recently were primarily being driven by decarbonisation. This includes data centres, power equipment, grid capacity, cooling systems, energy storage, energy infrastructure and efficient computing architectures. In that sense, AI reinforces a wider shift in markets: from operating expenditure to capital expenditure, from digital to physical infrastructure and a renewed demand driver for large parts of the market, including semiconductors, industrials, materials and utilities.

What we attempt to answer from an investment perspective is not whether AI could accelerate or slow the transition. It is how AI changes the shape of demand, the allocation of capital and the sources of competitive advantage across the real economy.

## Climate at the Crossroads of Three Megatrends

Sources: BloombergNEF, Energy Transition Investment Trends 2026; McKinsey & Company; International Energy Agency; Franklin Templeton Global Investments estimates.

Environmental spend increasingly overlaps with three structural shifts reshaping the next decade. In this three-part series, we examine the interaction of some of these trends and their implications for climate change investing.

### Power Demand

The first and most obvious implication of the AI buildout is seen in power. Training and inference capacity is turning data centres into large, fast-growing loads on power systems. The International Energy Agency (IEA) estimates that data centre electricity consumption was around 415 TWh in 2024 and projects it to reach around 945 TWh by 2030 (around 3% of total electricity demand).<sup>1</sup> The need for greater capacity and speed of capacity is making grid integration more challenging.

The investment implication is that the bottleneck is not only the absolute amount of power required. It is also about location, speed and reliability. Data centres tend to arrive in clusters, require very high uptime and can be developed faster than transmission networks, permitting systems and power-equipment supply chains. This can create local power-market tightness even where national electricity systems look adequately supplied. For example, new power pricing contracts for wind and solar projects in the United States continue to reach multi-year highs on the back of this squeeze, despite ongoing political uncertainty. For investors, the relevant opportunity set therefore extends into power generation, grid equipment, energy storage, power semiconductors, thermal management, monitoring software, backup power and energy efficiency. The purchase decision for renewable energy today is increasingly driven by power availability and speed-to-grid in addition to decarbonisation efforts.

A new wind farm, solar project or battery installation can, of course, reduce emissions, but it can also provide quick, low-cost capacity at a time when grid constraints mean that large-scale gas or nuclear alternatives remain years away. For hyperscalers<sup>2</sup> installing data centres, access to clean and reliable power by means of renewables is a way to satisfy an urgent need for power. For governments, it can help solve grid buildout challenges while decarbonising supply and creating 'green' jobs. For investors, it is a new growth opportunity for clean renewable power, despite fading subsidies in the developed world.

### Extraordinary Energy Efficiency

AI raises electricity demand, but the efficiency picture is more nuanced than the headline power numbers suggest. Google reported that its data-centre energy emissions fell 12% in 2024 despite data-centre electricity consumption rising

27% year-on-year, largely reflecting the impact of more than 25 previously contracted clean-energy projects coming online.<sup>3</sup>The company also claims its data centres now deliver over six times more computing power per unit of electricity than they did five years ago. Rapidly improving GPU<sup>4</sup>and related software architecture has been a key driver here-with several-fold performance increases from a similar power level with each new architecture. At the same time, part of this improvement has been driven by AI itself, for example the Gemini-powered 'AlphaEvolve'<sup>5</sup>agent has helped to reduce training time by around 1% and saved 0.7% fleet-wide compute through improved task scheduling. While these figures should not be treated as evidence that AI 'solves' its own footprint, they do show that more efficient computing architectures and infrastructure efficiency are lowering the cost of compute, enabling more AI use cases with potential savings on the other side. As AI becomes more efficient in its power usage, a scenario of system-wide energy savings through end-user adoption becomes much more plausible.

The specific impacts of AI remain up for debate and are clearly very broad in nature. However, early evidence shows that its use can improve predictive maintenance, building energy management, industrial process control, logistics, grid balancing and power-market forecasting. In each case, the value proposition is similar: better use of physical assets, lower waste and faster decision-making. The transition has often been framed as a need to build more low-carbon assets to replace older and less efficient ones. While that remains crucial, using existing assets more intelligently can often deliver more immediate decarbonisation. Example of how a wind turbine could change its pitch or yaw control? The IEA has highlighted AI's potential to optimise energy systems and support innovation in new energy technologies, though the scale of that benefit will depend on implementation, data quality and incentives.

## **Lower Pricing Drives Higher Demand**

Clearly, while efforts are being made to reduce environmental impact, the AI buildout is driving new-and incremental-power demand. More precisely, despite rapid declines in compute costs through energy efficiency improvements, usage of it is rising as a result. This elasticity of demand (often referred to as Jevon's Paradox<sup>6</sup>) is not to be ignored or dismissed, nor is it new. Similar debates have accompanied LEDs, electric vehicles and solar power. Falling unit costs generally increase adoption and as a result, boost total demand. The enabling of higher air conditioning usage via a more rapid solar build-out has not undermined the case for solar. In the same spirit, rising AI usage should not be a reason to dismiss AI infrastructure. Our focus is on whether a business is enabling the next unit of demand with improved efficiency, lower carbon intensity and defensible economics.

## **Compute: Cost vs Usage (2010-2030)**

Sources: Cisco Global Cloud Index, Fiber Dan, McKinsey. There is no assurance that any estimate, forecast or projection will be realized. Data Sources & Forecast Assumptions: Cost: Epoch AI GPU Price-Performance - FLOP/s per \$ doubles every ~2.5 years (forecast continues trend). Usage: Cisco Global Cloud Index, Fiber Dan - actual data centre traffic (ZB/year). Forecast: McKinsey projects compute workloads nearly triple by 2030 (AI + non-AI combined). Solid lines = Historical data | Dashed lines = Forecast (2026-2030).

## **A Broad Opportunity Set**

This leads directly to stock implications. AI should not be treated only as a semiconductor opportunity, nor simply as a software risk. A data centre is an industrial project. It requires advanced chips, custom accelerators, memory, substrates, specialist manufacturing equipment, pumps, compressors, power electronics, transformers, switchgear, cooling systems, renewable power and storage. Industrial companies like France's Saint-Gobain are hardly thought of by investors as AI beneficiaries, but are in fact seeing substantial growth in business segments providing liquid cooling solutions, HVAC insulation<sup>7</sup>and other materials that optimize airflow management within data centres, while also providing exterior "cool roof" solutions to reflect solar radiation and reduce building surface temperatures to radically lower the facilities' energy load.

The market may currently focus on the platform companies and chip designers, but the supply chain is much wider and many of the ingredients for building these massive capital projects will also decarbonise the electricity supply, driving significant growth opportunities for many of our companies. For example, UK utility operator SSE benefits greatly by providing grid-management tools, inverters, battery systems, high-voltage equipment and energy efficiency technologies, which all sit at the intersection of AI and the energy transition.

At the same time, AI presents disruption risk to many investments. Portfolio companies must be tested not only for exposure to AI-related demand, but also for vulnerability to automation, software substitution and changing customer behaviour. In some industries, AI may expand the addressable market. In others, it may compress margins, reduce

labour intensity or shift value up or down the supply chain. We must remain vigilant in monitoring emerging changes in competitive structure and other profit-drivers.

In terms of portfolio construction, the message is one of broadening out. AI remains a dominant market theme, but the opportunity set should broaden beyond the narrow group of AI-adjacent leaders that have driven recent market performance. As industrial and climate-linked sectors work through the bottom of their cycles, we expect a more supportive economic environment to allow a wider range of companies to benefit from capex tied to electrification, grid resilience and efficiency. We think it sensible for investors to maintain exposure to genuine AI beneficiaries but avoid allowing one narrative to create excessive concentration.

The AI data-centre buildout extends well beyond semiconductors, creating investment opportunities across the physical infrastructure required to deliver reliable, efficient and lower-carbon compute.

Sources: Company disclosures, industry sources, author analysis. For illustrative purposes only.

## **AI: Reshaping Climate Transition Economics**

AI and climate are often discussed as a tension between digital growth and environmental costs. In our view that framing is incomplete. AI is both a new source of power demand and a tool for improving the efficiency of our current infrastructure. The investment opportunity lies in identifying the companies that can supply more computing power at lower cost, store it more effectively and utilise it more intelligently. In that sense, AI does not sit outside the climate transition but rather as one of the key forces reshaping its economics.

## **Endnotes**

Source: "Energy and AI." International Energy Agency (IEA). 10 April 2025.

Hyperscalers are mega-scale cloud and technology companies that own and operate vast data centre networks and can expand computing capacity on a global scale.

Source: Google 2025 Environmental Report.

A GPU (Graphics Processing Unit) is a specialized computer processor designed to perform many calculations simultaneously.

AlphaEvolve is a Gemini-powered AI coding agent developed by Google DeepMind to discover and optimize algorithms automatically.

Jevons' Paradox is the idea that when a technology becomes more efficient at using a resource, the total consumption of that resource can increase rather than decrease. The concept comes from economist William Stanley Jevons, who observed in 1865 that improvements in the efficiency of coal-fired steam engines made coal cheaper and more useful, which ultimately led to greater overall coal consumption, not less.

HVAC insulation is insulation used on Heating, Ventilation, and Air Conditioning (HVAC) systems to reduce heat transfer, improve energy efficiency, prevent condensation, and minimize noise.

### **WHAT ARE THE RISKS?**

All investments involve risks, including possible loss of principal.

Equity securities are subject to price fluctuation and possible loss of principal.

Small- and mid-cap stocks involve greater risks and volatility than large-cap stocks.

Investment strategies which incorporate the identification of thematic investment opportunities, and their performance, may be negatively impacted if the investment manager does not correctly identify such opportunities or if the theme develops in an unexpected manner. Focusing investments in the health care, information technology (IT) and/or technology-related industries carries much greater risks of adverse developments and price movements in such industries than a strategy that invests in a wider variety of industries.

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